

D 130226

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Name.....

Reg. No.....

**FIFTH SEMESTER (CBCSS—U.G.) DEGREE EXAMINATION
NOVEMBER 2025**

Mathematics

MTS 5B 05—ABSTRACT ALGEBRA

(2020 Syllabus)

Time : Two Hours and a Half

Maximum : 80 Marks

Section A*Answer any number of questions.**Each question carries 2 marks.**Ceiling is 25.*

1. Define Euler's totient function ϕ hence find $\phi(10)$.
2. Check whether the relation on $\mathbb{R}$ defined by $a \sim b$ if $a \leq b$, where $a, b \in \mathbb{R}$ is an equivalence relation.
3. Write the permutation :

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ 5 & 2 & 7 & 6 & 3 & 8 & 1 & 4 \end{pmatrix}$$

in S_8 as disjoint product of cycles and as a product of transpositions.

4. Show that in a group G and $a, b \in G$, each of the equations $ax = b$ and $xa = b$ has a unique solution.
5. Write the addition table for $\mathbb{Z}_8$.
6. Let G be a group and $a \in G$. Then show that $\langle a \rangle$ is a sub-group of G .
7. Show that any group of prime order is cyclic.
8. Let $\phi: G_1 \rightarrow G_2$ be an isomorphism of groups. If a has order n in G_1 , then show that $\phi(a)$ also has order n in G_2 .

Turn over

9. Show that the set of all even permutations of S_n is a sub-group of S_n .
10. Give the sub-group diagram of Z_{60} .
11. Find the order of the permutation $(1, 2, 3) (2, 4, 3, 5) (1, 3, 2)$ in S_5 .
12. Let $\phi: G_1 \rightarrow G_2$ be a group isomorphism. Show that $(\phi(a))^{-1} = \phi(a^{-1})$ for all $a \in G$.
13. Show that sub-groups of index two are normal.
14. Let G be a group. Prove that $\text{Aut}(G)$ is a group under composition of functions.
15. Define units in a ring. What are the units in Z ?

Section B

Answer any number of questions.

Each question carries 5 marks.

Ceiling is 35.

16. Let $S = \mathbb{R} - \{-1\}$. Define $*$ on S by $a * b = a + b + ab$. Show that $(S, *)$ is a group.
17. Prove that every permutation in S_n can be written as a product of disjoint cycles.
18. Let G be a group, and let H and K be sub-groups of G . If $h^{-1}kh \in K$ for all $h \in H$ and $k \in K$, then HK is a subgroup of G .
19. Let G be an infinite cyclic group. Show that $G \cong \mathbb{Z}$.
20. Let G be a group, and let $a, b \in G$ be elements such that $ab = ba$. If the orders of a and b are relatively prime, then $o(ab) = o(a) o(b)$.
21. Show that any finite integral domain is a field.
22. Let $\phi: G_1 \rightarrow G_2$ be a group homomorphism. If H_1 is a sub-group of G_1 , then show that $\phi(H_1)$ is a sub-group of G_2 .
23. State and prove First Isomorphism Theorem.

Section C

*Answer any **two** questions.*

Each question carries 10 marks.

Maximum 20 marks.

24. Let S be any set and σ, τ are disjoint cycles in $\text{Sym}(S)$. Show that $\sigma\tau = \tau\sigma$.
25. Show that every group is isomorphic to a group of permutations.
26. Let H be a sub-group of the finite group G . Show that the order of H is a divisor of order of G .
27. State and prove Second Isomorphism Theorem.